A

Synopsis

on

CAMPUS PLACEMENT PORT

in partial fulfillment of the requirement for the degree

of

Bachelor of Technology

In

COMPUTER SCIENCE AND ENGINEERING

Submitted by

Yashvi Dixit (2301330100233)

Vishal Mavi (2301330100223)

Mansi Yadav (2301330120057)

Vishal Rai (2301330100228)

Under the supervision of

Mr. Rajat,

Ms. Anuradha,

Ms. Shivani, Ms.

Gourav, Ms. Roshni

(Designation CSE)

NOIDA INSTITUTE OF ENGINEERING AND TECHNOLOGY

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INTRODUCTION

Campus placement is a critical process in engineering institutions that connects students with potential employers. However, many institutions still rely on manual processes, spreadsheets, and email-based communication to manage recruitment activities. This often results in inefficiencies, delays, data inconsistencies, and lack of transparency.

The Real-Time Campus Placement Management System is a web-based application designed to automate and streamline the placement process. The system provides a centralized platform where students can create profiles, upload resumes, and apply for job opportunities. Recruiters can post job openings, view eligible candidates, and update selection status. The placement cell can monitor and manage the entire recruitment workflow.

Technology Used

- Frontend : HTML, CSS, JavaScript/React
- Backend : Node.js/ PHP
- Database : MySQL
- Authentication : Role-based login System

This project falls under the domain of **Web Application Development and Database Management Systems (DBMS).**

EXISTING SYSTEM

Current Scenario:

Features	**Existing System**	**Limitations**
Student Records	Maintained in Excel sheets	Data redundancy
Company Communication	Email-based	Delayed updates
Application Tracking	Manual	No real-time status
Shortlisting	Manual filtering	Time-consuming

Drawbacks:

- No centralized system
- Lack of transparency
- High manual effort
- Error-prone data management
- Difficult to track application status.

Problem Statement

Campus placement is a critical process in higher educational institutions, yet many colleges still depend on manual or semi-digital methods such as spreadsheets, emails, and notice boards to manage recruitment activities. These traditional approaches lack integration, automation, and real-time coordination, resulting in inefficiencies and operational challenges.

Students often face difficulties in accessing updated information about job opportunities, eligibility criteria, and recruitment schedules. There is no proper mechanism for tracking application status in real time, which creates confusion and uncertainty. Placement cells struggle with managing large volumes of student data, verifying eligibility manually, and maintaining accurate records of applications and selections. Recruiters, on the other hand, find it time-consuming to filter and shortlist suitable candidates due to unstructured or scattered data.

The absence of a centralized system leads to communication gaps, data redundancy, delayed updates, and increased chances of human error. Additionally, the lack of transparency in the recruitment workflow reduces accountability and efficiency.

Therefore, there is a strong need to develop a centralized, secure, and real-time campus placement management system that automates student registration, job postings, resume management, eligibility verification, and application tracking. The proposed system aims to streamline the entire recruitment process, improve transparency, enhance coordination among stakeholders, and ensure efficient data management within the institution.

Proposed Methodology

System Architecture

The system follows a three-tier architecture:

1. Presentation Layer (Frontend) - Handles user interaction through web interfaces.
2. Application Layer (Backend) – Processes requests, business logic, and authentication.
3. Application Layer (Backend) – Processes requests, business logic, and authentication.

This layered approach improves scalability, security, and maintainability.

Database Design

The database is relational and structured to reduce redundancy using normalization techniques.

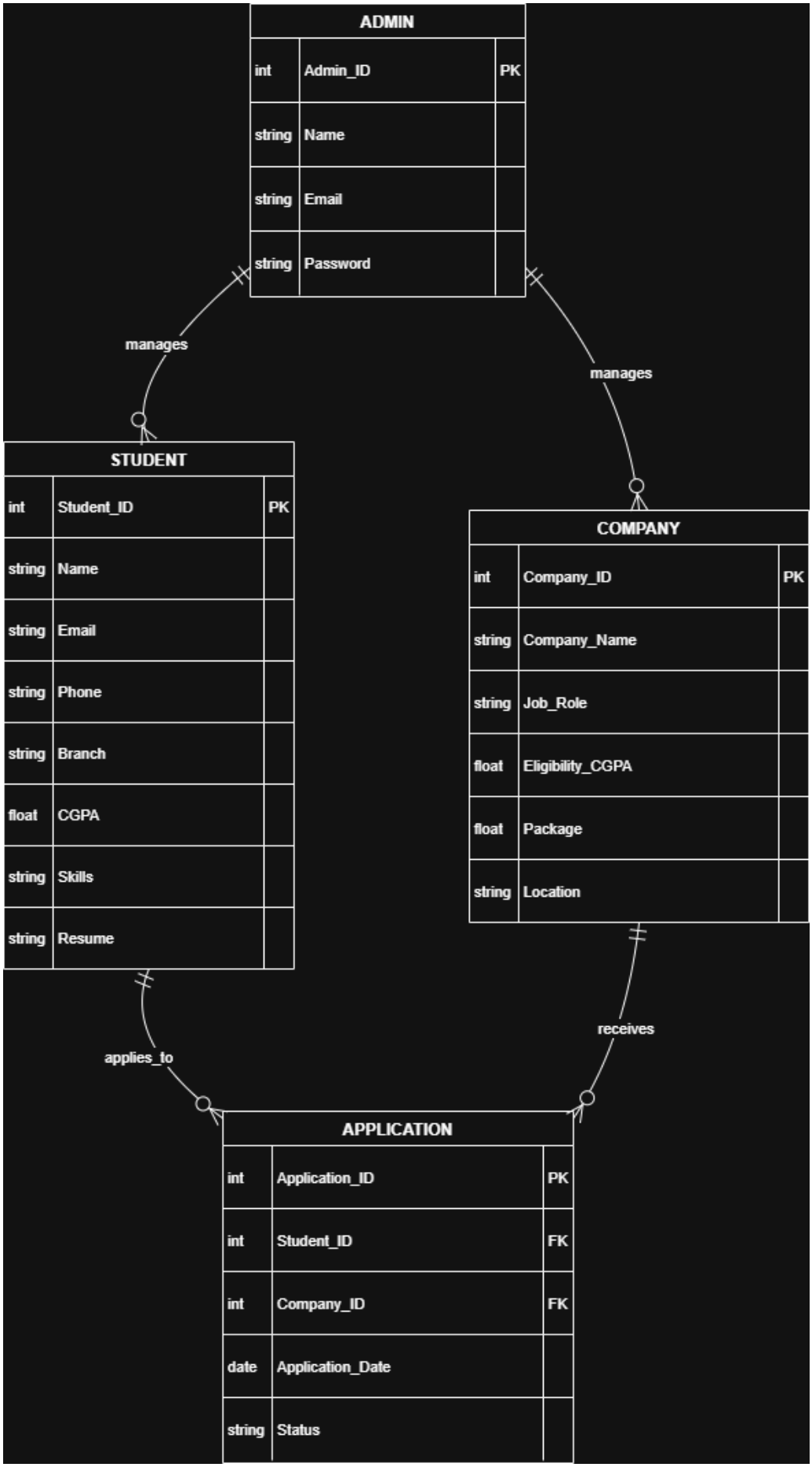
Main Tables:

- Student Table – Stores personal and academic details.
- Company Table – Stores job postings and eligibility criteria.
- Application Table – Tracks job applications and status updates.
- Admin Table – Stores system control data.

Primary and foreign keys ensure proper relationships between entities.

Security Mechanisms

- Role-based authentication (Student/Admin/Recruiter)
- Password encryption
- Input validation to prevent SQL injection
- Secure file upload handling for resumes.
  ER- Diagaram: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int Admin_ID PK
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string Email
string Password
manages
manages
STUDENT
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string Email
string Phone
string Branch
float CGPA
string Skills
string Resume
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string Company_Name
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float Eligibility_CGPA
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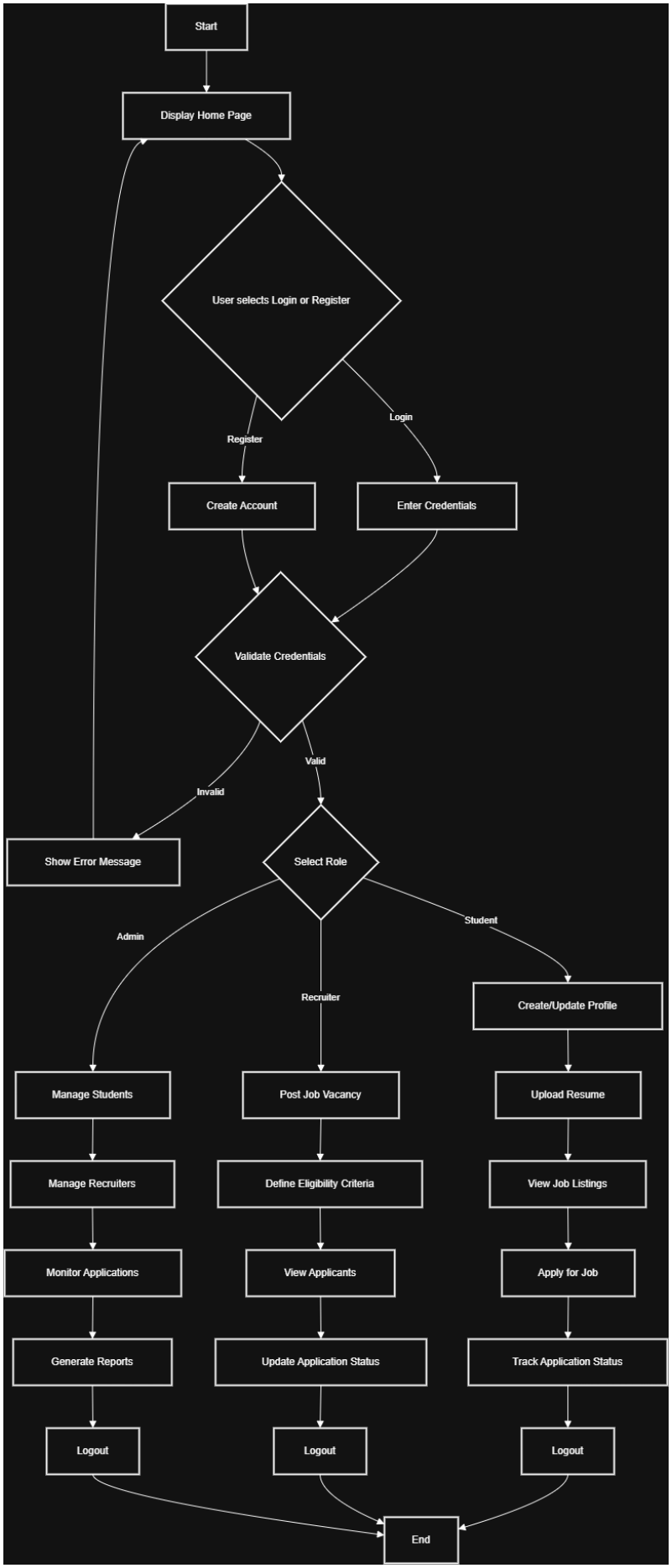
Start
Display Home Page
User selects Login or Register
Register
Login
Create Account
Enter Credentials
Validate Credentials
Valid
Invalid
Show Error Message
Select Role
Admin
Recruiter
Student
Create/Update Profile
Manage Students
Post Job Vacancy
Upload Resume
Manage Recruiters
Define Eligibility Criteria
View Job Listings
Monitor Applications
View Applicants
Apply for Job
Generate Reports
Update Application Status
Track Application Status
Logout
Logout
Logout
End

Feasibility Study

Technical Feasibility

The system uses standard web technologies that are widely supported. Developers have access to required tools and frameworks.

Economic Feasibility

Since open-source software and minimal hardware are used, development and deployment costs are low.

Operational Feasibility

The interface is simple and intuitive. Training requirements are minimal.

Time Feasibility

The project can be completed within one academic semester.

Facilities Required

Hardware Requirements:

- Computer/Laptop
- Minimum 4GB RAM
- Minimum 4GB RAM

Software Requirements:

- Windows/Linux Operating System
- VS Code / Any IDE
- MySQL Database Server
- Web Browser

Conclusion

The Real-Time Campus Placement Management System provides a structured and efficient solution for managing recruitment activities in educational institutions. By automating student registration, job postings, resume management, and application tracking, the system reduces manual effort and improves transparency. It enhances coordination among students, recruiters, and placement administrators, ensuring a smooth and organized placement process.